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CONNECTION MANAGEMENT OVER MULTIPLE NETWORKS

Abstract

One example may include receiving data, via a client device, over a first connection of a first network, sending a request, via the client device, to access a second network detected by the client device while the client device is communicating over the first network, responsive to receiving a captive portal from the second network, forwarding, via the client device, authentication information over a second connection to the second network, and receiving additional data, by the client device, over the first connection provided by the first network and a second connection provided by the second network after the authentication information is approved.

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Background/Summary

FIELD OF INVENTION

[0001] The present invention relates to secured communication and, in particular, to the use of secured communication over restricted networks. Specifically, an apparatus and method are disclosed for enabling secure communication (such as VPN) over a network with restricted access when secure communication (such as VPN) has previously occurred over another network.

BACKGROUND

[0002] While connected to a secure computing environment, users are typically protected from unauthorized receipt of their communication. Furthermore, chances of eavesdropping, destruction, or other adverse actions may be prevented by minimizing the chance of infection via spyware or malware. Technology users thus prefer to perform computer interaction on an enterprise network (or a trusted network) to ensure computer security.

[0003] It is not possible, however, to always confine oneself to a secure environment. Users are often in places where they require Internet access, but access is only available via a public network that is not secure. More specifically, users may need to connect a laptop and/or mobile phone to the Internet, may require various levels of security in their computation and communication tasks, and may find that the only practical way to communicate with the Internet is via a publicly available server. Connecting to the Internet in such an environment exposes the user to eavesdropping, data destruction, hacking etc. Another danger, however, also exists. When users are on their secure networks, firewall and security software often protects them from being able to communicate with sites that can potentially cause harm. Outside of the secure network, there might not be a “gatekeeper” to prevent a user from engaging in harmful site access. Furthermore, a user may access an endpoint with malware present, and such malware can subsequently gain access to enterprise resources.

[0004] Access to unsecured networks is often simple to achieve, since such networks are typically wireless. It is very common for such a network to be a local area network (LAN), and access to the LAN is provided via a wireless standard such as Wi-Fi. An access point provides a physical interface at which a user gains access to the LAN. The LAN may then transmit data to and receive data from a wide area network (WAN) such as the Internet.

[0005] Because wireless access has become so simple for the average layperson to achieve, wireless communication techniques are commonly used with a wide variety of mobile terminals, including smartphones, electronic whiteboards, projectors, laptops, etc. Wireless communication is performed by gaining entrance to a network using an access point. Because it is often not possible to visually confirm who is gaining access to wireless network, security relating to wireless communication typically needs to be more enhanced than in wired communication. Thus, before the terminal is connected to a wireless access point, the terminal may first be authenticated. Furthermore, in many wireless applications, data that is transmitted and received is encrypted. Another way to protect the wireless network is to limit access to terminals that have MAC addresses that are registered in advance. When the terminal attempts to connect to an access point,

the MAC address of the terminal is compared to a list of MAC addresses. Connection is only permitted if the terminal's MAC address is on the list.

[0006] While the use of MAC addresses as a security tool works well, such a strategy also creates substantial limitations. In particular, a user with a legitimate right to be on a private network may be denied access because the MAC address of the user's terminal has not been preregistered. As another example, a guest having a smart phone, laptop, or tablet may be invited into a conference room to exchange information or have a discussion using the guest's electronic device. If the MAC address of the device has not been preregistered, use of the device on the private network is not permitted.

[0007] As a further example, a substantial number of businesses operate private networks in their respective physical locations. Such offerings have numerous advantages. For example, many buildings are constructed of materials that interfere with the cellular communication of a user that is inside the building. An individual shopping in a retail store may thus be unable to communicate with their cellular network. A business may wish to offer their own private wireless network to all shoppers as part of providing good customer service. If a shopper has good wireless communication during their shopping experience, the shopper is incentivized to stay in the store for a longer period of time, and to make additional purchases. As another example, current Wi-Fi technology allows a shopper on a Wi-Fi network to be position tracked as they move about the store. In this manner, it is possible to obtain intelligence regarding shopper behavior.

[0008] Providing the public with access to a private network, while advantageous, is not without its risks. For example, the owner of a private network typically does not wish to give every member of the public unrestricted access to their network. At a minimum, it may be desirable to insist that each network user agree on terms or conditions for network access in order to limit liability. Also, it may be very desirable to advise potential network users of rules they are required to follow while using the network in order to maintain network integrity. In other situations, prior to granting network access, it may be desirable to obtain some form of identification and/or password so that the network owner has information regarding who is on their network. In other situations, network owners may desire greater quantities of information to ensure security and/or enable future advertising or marketing to network users. In a further situation, network owners may wish to limit network access to individuals that are actually making purchases within their establishment (by providing them a password, for example, after making a purchase). In yet further situations, network owners may desire to be paid for network access before network access is granted.

[0009] One way to achieve the above goals is by providing limited accessibility to a network. Such a network may be referred to as "closed." The limited accessibility allows a user terminal to request authentication from the network. Once authentication has been completed, the user terminal is then permitted greater (or possibly total) access to the network.

[0010] One common way to provide authentication to a user terminal seeking network access is via a captive portal. Wikipedia defines a captive portal as follows: . . . a web page accessed with a web browser that is displayed to newly connected users of a Wi-Fi network before they are granted broader access to network resources. Captive portals are commonly used to present a landing or log-in page which may require authentication, payment, acceptance of an end-user license agreement, acceptable use policy, survey completion, or other valid credentials that both the host and user agree to adhere by. Captive portals are used for a broad range of mobile and pedestrian broadband services—including cable and commercially provided Wi-Fi and home hotspots. A captive portal can also be used to provide access to enterprise or residential wired networks, such as apartment houses, hotel rooms, and business centers.

[0011] Network resources are thus withheld from a user terminal requesting network access until the requirements of the captive portal have been satisfied. The captive portal may be transmitted to a prospective network user in response to a request for network access. An example of a request for network access is an HTTP request for a web page from a requesting terminal. A captive portal

authentication server evaluates the request for network resources and determines if the request has come from a source that has already been granted (full) network access. If not, the authentication server responds to the HTTP request with either the contents of the captive portal page, or an HTTP Redirect response which sends the HTTP client to the URL of the captive portal page where it is hosted. Once required information has been provided to the authentication server in response to the captive portal, widespread network access is then granted to the requesting terminal.

[0012] The network for which access is granted by the captive portal may be referred to as a closed network, an initially closed network, or a captive network.

[0013] Captive portals control network access. They do not, however, address the issue of secure communication. In fact, once access to a network guarded by a captive portal is granted, a user is often communicating in a non-secure environment, and risk of security breach may be high.

[0014] Generally speaking, one method of allowing users to have secure communication over a network is through the use of a secure communications channel. An example of a secure communications channel is a Virtual Private Network (VPN). Various types of VPNs are available. A corporate VPN enables access to the internal resources (intranet) of an enterprise. A private VPN permits access to Internet resources. When data is sent over a public network using a VPN, eavesdropping of the data is prevented.

[0015] In order to communicate over the Internet, data segments from layer-4 in the OSI model (Transport) are divided into packets. An IP packet is then formed by adding header information to the data unit received from layer-4. The resulting IP packet comprises an IP header and an IP payload (where the data is located). In the header, address information is included such as the source of the data and the destination of the data. The source of the data is included in the header so that once the packet is received at its destination, a response can be transmitted to the data source. The IP packets form IP traffic (also known as layer-3 traffic). The communications format may also be referred to as TCP/IP.

[0016] In order to exchange data over a VPN, the IP traffic is received by the VPN software (i.e. the VPN client). The VPN software a) optionally encrypts each IP packet b) encapsulates each IP packet with a new header. Again, step a) is optional and step b) is not optional. The new header includes the VPN server as the destination address and the client as the source address. Once the encapsulated IP traffic is received by the VPN server, the new header is removed, encrypted data (if any) is decrypted, and the IP packet is delivered to its intended destination.

[0017] When referring to a VPN, application data and its associated header (for example an IP packet) is considered to be “encapsulated” when a further header that includes the destination address of a VPN server is attached to the application data (and associated header). When referring to the transmission of application data (and its associated header) outside of a VPN, application data is considered to be “unencapsulated” when it is not encapsulated (i.e. a header that includes the destination address of a VPN server is not attached to the application data and its associated header) and “decapsulated” when encapsulation is removed. Put another way, unencapsulated application data (and its associated header), for example an IP packet, is 1) encapsulated by a VPN client, 2) the encapsulated application data and header is transmitted to a VPN server, 3) the VPN server decapsulates the encapsulated application data, and 4) the decapsulated application data is forwarded to a final destination.

[0018] In the above example the encapsulated data may or may not be encrypted.

SUMMARY OF THE INVENTION

[0019] An example method of operation may include forwarding a request sent outside a VPN server, via a client device, to access a second communication network detected by the client device, and the client device is communicating with the VPN server over a first communication network, responsive to receiving a captive portal, forwarding, via the client device, authentication information to obtain access to the second communication network, and the authentication information is not forwarded to the VPN server, and receiving data, by the client device, from a

remote server over a bonded connection including a first connection provided by the first communication network bonded with a second connection provided by the second communication network to form the bonded connection.

[0020] Another example embodiment may include a client device including a processor configured to forward a request sent outside a VPN server to access a second communication network detected by the client device, and the client device is communicating with the VPN server over a first communication network, responsive to receiving a captive portal, forward authentication information to obtain access to the second communication network, wherein the authentication information is not forwarded to the VPN server, and receive data from a remote server over a bonded connection comprising a first connection provided by the first communication network bonded with a second connection provided by the second communication network to form the bonded connection.

[0021] Another example embodiment may include a non-transitory computer readable storage medium configured to store instructions that when executed cause a processor to perform forwarding a request sent outside a VPN server, via a client device, to access a second communication network detected by the client device, and the client device is communicating with the VPN server over a first communication network, responsive to receiving a captive portal, forwarding, via the client device, authentication information to obtain access to the second communication network, wherein the authentication information is not forwarded to the VPN server, and receiving data, by the client device, from a remote server over a bonded connection comprising a first connection provided by the first communication network bonded with a second connection provided by the second communication network to form the bonded connection.

[0022] Another example embodiment may include receiving data, via a client device, from a virtual private network (VPN) server over a first connection of a first network, sending a request sent outside the VPN server, via the client device, to access a second network detected by the client device while the client device is communicating with the VPN server over the first network, responsive to receiving a captive portal from the second network, forwarding, via the client device, authentication information over a second connection to the second network, and receiving additional data, by the client device, from the VPN server over the first connection provided by the first network and a second connection provided by the second network after the authentication information is approved.

[0023] Another example embodiment may include a process including receiving data, via a client device, over a first connection of a first network, sending a request, via the client device, to access a second network detected by the client device while the client device is communicating over the first network, responsive to receiving a captive portal from the second network, forwarding, via the client device, authentication information over a second connection to the second network, and receiving additional data, by the client device, over the first connection provided by the first network and a second connection provided by the second network after the authentication information is approved.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0024] FIG. 1 is a block diagram that illustrates communication with a remote server via a captive portal.

[0025] FIG. 2 is a block diagram that illustrates communication via a secure channel (such as a VPN) and over a captive network in accordance with an exemplary embodiment of the present invention.

[0026] FIG. 3 is a block diagram that further illustrates communication in accordance with FIG. 2,

and that also illustrates operation of a VPN client, in accordance with an exemplary embodiment of the present invention.

[0027] FIG. 4 is a flowchart diagram that illustrates operation of an exemplary embodiment of the present invention.

[0028] FIG. 5 is a further flowchart diagram that illustrates operation of an exemplary embodiment of the present invention.

DETAILED DESCRIPTION

[0029] Today it is well-known that substantial amounts of communication take place over the Internet. Such communication can include (for example) voice communication, video transmission, and written correspondence (e.g., email, chat, instant messaging, Tweets, etc.). The use of electronic devices has become so prevalent that today almost all adults in the United States own some sort of mobile phone. In addition to a mobile phone, some users may carry a further device such as a laptop or tablet.

[0030] To meet increasing demand for Internet access, public access to the Internet is widely available. Facilities such as shopping centers, malls, stores, restaurants, office buildings, and public transportation may offer open connections to the Internet. These connections may often be available without paying a fee.

[0031] While it is beneficial for users to have widespread access the Internet, there are also challenges because such connections may be unsecure. For example, a user communicating via an unsecured Internet connection may transmit signal packets that are easily intercepted and/or viewed by others. Transmission of passwords, credit cards, and other sensitive information may thus be readily viewed by unintended parties.

[0032] FIG. 1 illustrates a user communicating with a network via a mechanism called a captive portal. Initially, system 100 illustrates that client 110 wishes to communicate with remote server 120. As shown, client 110 may be, for example, a mobile communications device that wirelessly communicates with network 130 via respective access point (AP) 131. Network 130 may be any type of network including a public network or a private network. As an example, network 130 may be a cellular communications network that is maintained by a cellular communications provider (such as Verizon, AT&T, Sprint, etc.). Network 130 is shown coupled to Internet 125 and communicating with Internet 125 through communication protocols that are well known to one of ordinary skill in the art. In one example, network 130 interfaces with Internet 125 via a fiber optic or Ethernet Internet connection.

[0033] While in one example network 130 is a public network, the term public network refers to a (publicly available) network of devices that is not limited to particular devices, and that are able to communicate with other devices on the network. Thus, in one example, network 130 is accessed without a password. Alternatively, network 130 may be a private network and any device communicating therewith must be authenticated before communication with network 130 is permitted. Furthermore, while one access point is shown in FIG. 1, it is understood that typically (but not always) multiple access points may be used to gain access to network 130. For example, if network 130 is a cellular network, access points may be located in numerous geographic locations. Alternatively, if network 130 is a LAN, the access point may be a wireless router. Network 130 may thus be a LAN such as a wireless home network, a combination of wired and/or wireless connections, and perhaps one or more intervening networks (such as a wide area network) so that access to Internet 125 may be obtained.

[0034] In the example above, a user may use client 110 for voice communication. Assume client 110 is a cell phone, such as a smart phone, that is capable of communication via a Voice over IP (VOIP) application. Client 110 communicates wirelessly with network 130 via respective access point 131 and a digitized form of the user's voice is then transmitted to Internet 125. From Internet 125, the data that represents the user's voice is transmitted to remote server 120. From remote server 120, the data may be transmitted to another user (not shown) so that voice communication

between the two users may occur.

[0035] In another embodiment, client **110** streams video from remote server **120**. Client **110** requests the video data from remote server **110** by transmitting a request through network **130** and Internet **125**. Remote server **120** responds to the request by transmitting video via Internet **125**, and back to network **130**, so that it is eventually received by client **110**.

[0036] In the example shown in FIG. **1**, although client **110** is communicating via network **130**, client **110** now wishes to communicate via private network **140**. In one example, there may be a communication “hand-off” from network **130** to private network **140**. If, however, network **140** has restricted access, such as the restriction that exists when a captive portal is in use, a “hand-off” to private network **140** may not be possible. For example, if client **110** starts to transmit data to network **140**, but access to network **140** is restricted, then network **140** may refuse to receive (and/or carry) the data transmitted by client **110**. Such data may be prohibited from reaching its intended destination. As a result, when data transmission (e.g. voice communication, streaming, etc.) initially occurs via network **130**, and attempt is made to transition the data transmission to network **140**, the data transmission may be interrupted and/or data may be irretrievably lost.

[0037] In another example, while communication initially occurs solely via network **130**, communication subsequently may be desired via network **130** and private network **140** simultaneously. Such simultaneous communication may be desirable, for example, in a multipath environment (“bonded” or a “bonded connection”) that exists so that there is greater bandwidth than what is obtained in single path environment. An example of a multipath communications environment is provided, for example, by the product Speedify (manufactured by Connectify, Philadelphia, PA). Other multipath communication systems are well known to one of ordinary skill in the art. See for example Hoekstra, US 2007/0115987 which is incorporated herein by reference.

[0038] In yet another example, communication switches from occurring over network **130** to occurring over network **140**, but for some period of time, communication over network **130** and network **140** is concurrent.

[0039] While client **110** is communicating with network **130**, the individual that is using client **110** enters a physical area where access to private network **140** is desired. An example of private network **140** may be, for example, a LAN that is accessed via a communication protocol such as Wi-Fi. In one example, client **110** begins to receive a strong signal from access point **141** that potentially allows access to private network **140**, while a signal for obtaining access to network **130** becomes either weak or unavailable. In such an example, it is desired to “hand-off” communication from network **130** to private network **140**. In another example, while communication initially occurs with a strong signal for accessing network **130**, at a subsequent time strong signals exist for communication with network **130** and private network **140** concurrently. With such concurrent strong signals, concurrent access (bonding) to network **130** and network **140** may be desirable.

[0040] Before client **110** is provided access to private network **140**, client **110** is desirably authenticated. Initially, client **110** is prohibited (or limited) from communicating with Internet **125** via private network **140**. As an initial step, client **110** attempts to communicate with (or polls) private network **140**. Such polling may be, for example, by making an HTTP request. This HTTP request is initially routed to captive portal authentication server **150**.

[0041] Alternatively, such polling may be in the form of unencrypted data. Alternatively, such polling may be in a form that is recognized by captive portal authentication server **150** (as opposed to, for example, encrypted data that server **250** is unable to recognize).

[0042] Client **110** subsequently evaluates the response received from private network **140** to determine whether a captive portal is in place. For example, client **110** can transmit to private network **140** the URL of a website with a known response. As an example, if client **110** sends to the Internet an HTTP request to the host “www.msftncsi.com” and a path of “/ncsi.txt”, the expected response is the text string “Microsoft NCSI”. If client **110** does not receive the expected response (to the HTTP request, for example), then client **110** concludes that private network **140** is a captive

network that is protected by a captive portal.

[0043] When a captive portal is detected, authentication with the captive portal may occur. How authentication will actually occur with a specific captive portal will depend upon the requirements of the captive portal. For example, one captive portal may simply require agreement to terms of use. Another captive portal may require providing identification information and/or an email address. In other scenarios, the captive portal may require a credit card (so that the credit card can be charged) in order to obtain access to the Internet via a captive network. Typically, captive portal authentication server **150** transmits to client **110** an authentication web page (or redirects client **110** to an authentication web page) so that client **110** can provide server **150** with whatever information is required to give client **110** full (or greater) access to network **140**. In other words, server **150** takes an action that causes client **110** to receive an authentication web page. The authentication web page specifically requests from client **110** required information to grant full (or greater) access to network **140** (name, email address, credit card, acknowledgement of terms, etc.).

[0044] Captive portal authentication server **150** receives one (or more) responses to its action, of providing an authentication page, from client **110**. If captive portal authentication server **150** receives the required response(s), then captive portal authentication server **150** records/stores sufficient information so that client **110** has full (or greater) access to network **140**. “Sufficient information” may include the MAC address of client **110** (for example). In one example, during future attempts of client **110** to communicate with Internet **125**, the MAC address recorded by authentication server **150** is compared with the MAC address of client **110**, and if the two match, then full access to network **140** (and thus Internet **125**) is granted. In one exemplary embodiment, full (or greater) access to network **140** is granted until such access expires (i.e. an expiration date has been reached, a predetermined amount of data has been exceeded, a credit card has expired, etc.). While access to network **140** is granted, client **110** may not be redirected to captive portal authentication server **150** when client **110** initiates communication with private network **140**. Once access expires, client **110** upon initiating communication with private network **140** is again redirected to captive portal authentication server **150** until access to private network **140** has again been granted.

[0045] In any event, once captive portal authentication is complete, client **110** is permitted to communicate with remote server **120** via private network **140** and Internet **125**. As previously explained, communication with remote server **120** via private network **140** and network **130** can be concurrent, or can occur in a serial manner. In other words, once authentication is complete, both private network **140** and network **130** may transfer data between client **110** and server **120**. Alternatively, network **130** may transfer data between client **110** and server **120**, and then network **140** may transfer data between client **110** and server **120**. In addition, during the “hand-off” from network **130** to network **140**, there may be overlap, i.e. concurrent data transfer by network **130** and network **140** (before data transfer by network **130** ceases and data transfer by network **140** continues). In this manner, loss of data during the “hand-off” may be prevented.

[0046] Generally speaking, the use of captive portals is well known in the art. However, difficulties may arise when using a captive portal in combination with a secure communications channel such as a VPN. Simply put, as explained above, when communication occurs via a VPN, the data is encrypted. Thus, if communication via VPN is occurring via network **130**, and communication suddenly switches to private network **140**, VPN data will be transmitted to private network **140** despite the fact that the source of the VPN data, client **110**, has not been authenticated. In other words, VPN data will be transmitted to private network **140**, a captive network, even though the requirements of the captive portal (that prevents access to private network **140**) have not been satisfied. Without performing authentication of client **110** via captive portal authentication server **150**, VPN data transmitted by client **110** will be denied access to private network **140**, and VPN data intended to be transmitted over network **140** will be prevented from reaching Internet **125**. To the extent the VPN relies on network **140** to communicate over Internet **125**, that communication

will fail.

[0047] VPNs are typically used in various situations, including: [0048] Normal VPNs use only one Internet connection at a time; and [0049] Multipath VPNs use multiple Internet connections concurrently if they are available.

[0050] If a regular single socket VPN simply attempts to send encrypted data to a captive portal (without client authentication) via a single Internet connection, connection is typically not possible

[0051] If a single socket VPN has multiple Internet connections available, one of which has a captive portal, connection of the VPN on the captive network may fail, or the VPN may be active on networks other than the captive network (the network associated with the captive portal). Access to the captive network by the VPN may then be difficult if not impossible, as the server associated with the captive portal will receive encapsulated data, and when the server's requirements for granting access to the captive network are not met (because it cannot process the encapsulated data), it will refuse to permit the VPN to communicate over the captive network. The access point associated with the captive network will then simply be a non-working hotspot.

[0052] In an exemplary embodiment of the present invention, when data is being transmitted to a remote server via a VPN (prior to communicating via a captive network), operation of an exemplary embodiment of the present invention may be achieved as illustrated, for example, by FIG. 2.

[0053] In FIG. 2, VPN communication (as an example of secured and/or encapsulated communication) initially occurs between client **210** and VPN server **235**. Client **210** may be a mobile communications device. Client **210** may be a secured client such as a VPN client. As shown, VPN communication occurs through network **230** via its respective access point **231**, through Internet **225**, and to VPN server **235**. Client **210** and VPN server **235** each include respective VPN clients for encrypting and decrypting (and/or encapsulating and decapsulating) data transmitted between them. After encrypted/encapsulated data is received by VPN server **235**, the data is decrypted/decapsulated and delivered to one or more of various servers depending upon the exemplary embodiment. In one exemplary embodiment, the decrypted/decapsulated data is delivered to remote server **220** via private network **245**. In another exemplary embodiment, the decrypted/decapsulated data is transmitted to remote server **260** via Internet **225**.

[0054] If the data is transmitted to remote server **220**, then when remote server **220** responds, remote server **220** provides data (e.g. streaming video) to VPN server **235** via private network **245**. The data is encrypted/encapsulated by VPN server **235** and is subsequently transmitted to client **210** via Internet **225** and network **230**.

[0055] If the data is transmitted to remote server **260**, then when remote server **260** responds, remote server **260** provides data (e.g. streaming video) to VPN server **235** via Internet **225**. The data is encrypted/encapsulated by VPN server **235** and is subsequently transmitted to client **210** via Internet **225**.

[0056] It is noted that: remote server **220** may be optional when remote server **260** is available, remote server **260** may be optional when remote server **220** is available, or remote server **220** and remote server **260** may both be present and available.

[0057] At some point, client **210** (e.g. a secured client) is in range of access point **241** (and thus private network **240**). For example, client **210** eventually receives a sufficiently strong signal from access point **241** for communication therewith. This may occur, for example, when a user that is physically carrying client **210** enters a building in which Wi-Fi access to the Internet is (conditionally) offered to the public. As a result, access to private network **240** is desired via access point **241**. Access to private network **240** may be in place of or in addition to access with network **230**. Client **210** signals (or transmits data to) private network **240** (to communicate therewith) via access point **241**, but client **210** is prohibited (or limited) from accessing Internet **225** because access to network **240** is controlled by a captive portal.

[0058] While client **210** is attempting to gain access (or full access) to Internet **225** via private

network **240**, client **210** continues to have VPN communication with VPN server **235** via access point **231**. As captive portal authentication server **250** is authenticating client **210** (or at least attempting authentication), client **210** and VPN server **235** continue to exchange encrypted/encapsulated data via network **230** and Internet **225**. As VPN server **235** is receiving encrypted/encapsulated data from client **210**, VPN server **235** is decrypting/decapsulating that data and either (or both) transmitting that data via private network **245** to remote server **220** or transmitting that data via Internet **225** to remote server **260**. Similarly, while captive portal authentication server **250** is attempting to authenticate client **210**, remote server **220** may transmit data back to VPN server **235** via private network **245** or remote server **260** may transmit data back to VPN server via Internet **225**. The data received by server **235** from server **220** and/or server **260** is encrypted/encapsulated and transmitted to client **210** via Internet **225** and network **230**.

[0059] Captive portal authentication server **250** thus receives a signal from client **210** (via network **240**) indicating that client **210** desires to (more) fully access network **240** (and to thereby access Internet **225**). The signal that server **250** receives from client **210** may be, for example, an HTTP request to receive a website page with known text. The URL described above (with reference to FIG. **1**) that responds with the text “Microsoft NCSI” is an example of an HTTP request that when properly processed by the Internet provides a known response. If private network **240** does not return to client **210** an expected response (for example to an HTTP request), client **210** now knows that private network **240** is protected by a captive portal (because instead of receiving the expected response, client **210** was blocked from receiving the expected response via the captive portal).

[0060] In response to client **210** signaling private network **240**, captive portal authentication server **250** attempts to authenticate client **210**. This may be accomplished, as explained above, by providing a login screen, requesting agreement to the terms of use, providing a credit card, etc. Client **210** then has an opportunity to provide captive portal authentication server **250** with the required response. The authentication process thus comprises the following exemplary steps:

[0061] Client **210** transmits a signal (e.g. HTTP request) to private network **240** to determine whether private network **240** is guarded by a captive portal; [0062] Captive portal authentication server **250** responds to client **210**'s signal by transmitting to client **210** that there is a need for captive portal authentication, or by redirecting the request to the captive portal; [0063] When client **210** does not receive an expected response to its signal (that was transmitted in step **1**), client **210** now knows that network **240** is guarded by a captive portal; [0064] Client **210** provides server **250** with the authentication information that server **250** requires (if it can).

[0065] Server **250** checks the information provided by client **210** to confirm that all required information has been provided.

[0066] Server **250** grants permission to client **210** to (more) fully access network **240** (for example by retaining the MAC address of client **210**).

[0067] Client **210** (more) fully accesses network **240** (and thus accesses Internet **225** via network **240**), and does not require authentication from authentication server **250** during future transmissions from client **210** in order to access Internet **225** via private network **240**.

[0068] In a further exemplary embodiment of the present invention, an alternative process of finding the captive portal in step **1** may be used. For example, in RFC 7710-“Captive-Portal Identification Using DHCP or Router Advertisements (RAs)” (incorporated by reference in its entirety), clients are able to find if a network has a captive portal, and what URL to call to connect to it, via checking a DHCP option on an IPv4 network, or via Router Advertisement on an IPv6 network.

[0069] While authentication to (more) fully access private network **240** is occurring, VPN communication between client **210** and VPN server **235** is occurring via network **230**. While logging into the captive portal, the VPN is still able to keep other traffic flowing, encapsulated and encrypted on other connections so traffic (calls, downloads, etc.) are not interrupted.

[0070] Client **210** may thus respond to the server **250** in a manner that causes server **250** to grant

(more) access to private network **240**. After client **210** returns the response required by authentication server **250** (i.e. step **4** above), authentication server **250** may record identification information associated with client **210** so that, during a subsequent time that client **210** desires access to private network **240**, that access can be provided based upon the fact that client **210** has already been identified. One manner of identifying client **210** for future recognition (and permission) is by storing the MAC address associated with client **210** in database **252**. Other identification information can also be stored in database **252** including, for example, a username, a password, a credit card number, etc. Database **252** can then be examined when client **210** is trying to access private network **240**, and access to private network **240** can be granted if the MAC address of client **210** is stored in database **252**.

[0071] Once client **210** has been authenticated, captive portal authentication server **250** allows client **210** to communicate with Internet **225** via private network **240**. Client **210** is thus able to transmit encrypted data to VPN server **235** via private network **240** and Internet **225**, for a) subsequent receipt by private network **245** and remote server **220**; or b) subsequent receipt by remote server **260** via Internet **260**. Any response (e.g. streaming of data) from remote server **220** may likewise be channeled over private network **245**, VPN server **235**, Internet **225** and private network **240** before reaching client **210**. Any response from remote server **260** may likewise be channeled over Internet **225**, VPN server **235**, Internet **225** and private network **240** before reaching client **210**.

[0072] In one exemplary embodiment, a VPN channel initially exists only via network **230**. After client **210** has been authenticated by captive portal authentication server **250**, the VPN channel is discontinued over network **230**, and the VPN channel is active over private network **240** (in order for the VPN channel to be connected with VPN server **235**). In another embodiment, a VPN channel (or VPN channels) is simultaneously active over network **230** and private network **240**. As a result of having a VPN channel (or multiple VPN channels) that is active over two (or more) networks, more bandwidth is available than over either network alone. In yet a further embodiment, the VPN channel is initially established (and is active) over network **230**, is then established (or another channel is established) over private network **240** (so that the VPN channel is active concurrently over network **230** and private network **240**), and a VPN channel is subsequently active only over private network **240** (the channel over network **230** is discontinued). In this manner, there is overlap in the “handoff” between the two networks to ensure that no data is lost in the transition between the two networks.

[0073] In the above manner, while encrypted data is being transmitted to server **235** via network **230**, one or more of the following steps are occurring: transmitting a signal to network **240** in order to gain access (or expecting that access is available, whether or not it is), identifying a captive portal, and responding to the captive portal to gain (more) access to network **240**.

[0074] Communication between client **210** and VPN server **235** via private network **240** is further explained with reference to FIG. **3**. In particular, FIG. **3** illustrates a VPN client **310** that may be resident, for example, within client **210**.

[0075] VPN client **310** includes driver **315** that receives data from one or more applications, inserts the data into a VPN channel, and then forwards the data over the VPN channel to Internet **225** (and subsequently to VPN server **235**). Driver **315** may be, for example, a TUN/TAP driver.

[0076] Data transmitted from driver **315** is received by routing module **320**. The purpose of routing module **320** is to determine whether the received data will be further encapsulated (for purposes of being transmitted via a VPN) and to which network the data will be transmitted. Thus, initially, during VPN communication, data is forwarded to encapsulation module **330** and subsequently transmitted to network **230**. From there, the data is transmitted via VPN to Internet **225** and subsequently to VPN server **235**.

[0077] At some point, access point **241** is detected and client **210** desires to communicate over network **240** (so that communication with VPN server **235** occurs over network **240**). Therefore,

captive portal detector **325** will determine via routing module **320** whether private network **240** is a captive portal. In some embodiments, captive portal detector **325** will cause routing module **320** to launch proxy **335** in order to communicate with private network **240**. Proxy **335** may allow a web browser to be opened for login to a captive portal, and may be situated in the VPN client. It is understood to one of ordinary skill in the art that some operating systems (iOS) rely on the use of a proxy to communicate with private network **240**, while other operating systems (i.e. Windows) do not normally need a proxy. Whether or not proxy **335** is needed, VPN client **310** communicates with private network **240** (through optional proxy **335**) to determine whether private network **240** is protected by a captive portal. In one embodiment, proxy **335** transmits a signal to private network **240** in order to see how private network **240** responds.

[0078] For purposes of clarity, in order to transmit VPN traffic over private network **240** after (or while) traffic (such as VPN traffic) is transmitted via network **230**, it is desirable for private network **240** to be permitted to transmit VPN traffic thereon. For example, if VPN traffic simply begins to be transmitted from encapsulation module **330** to private network **240** (with a captive portal in place), private network **240** will simply ignore the encapsulated data being transmitted from encapsulation module **330** because the captive portal has not given permission for data (including encapsulated data) to be sent over private network **240** to Internet **225**. Therefore, as private network **240** is coupled to captive portal authentication server **250**, proxy **335** transmits to private network **240** data that has not been encapsulated in order to gain access to private network **240**. An example of data transmitted to private network **240** is the HTTP request described above. If in response to the HTTP request an expected response is not received, then captive portal detector **325** determines that proxy **335** is in communication with a captive network that is protected by a captive portal. Captive portal detector **325** then instructs proxy **335** to interact with captive portal authentication server **250** until client **210** has been authenticated. Once captive portal detector **325** determines that client **210** has been authenticated, a VPN channel to Internet **225** may now be active via private network **240**.

[0079] FIG. 4 and FIG. 5 are flowchart diagrams that illustrate operation of an exemplary embodiment of the present invention. FIG. 4 and FIG. 5 are each two flowchart diagrams. On the left side of each figure, as an example, an indication is provided of which network is currently carrying an active VPN channel. For example, as shown in FIG. 5, there may be a time period during which a VPN channel is active over both the first exemplary network and the second exemplary network. Alternatively, and not shown, there may not be an overlap time, i.e., the VPN channel may cease to be active on one network and simultaneously begin to be active on another network. In another exemplary embodiment, as shown in FIG. 5, when the VPN channel becomes active on the second network, termination of the VPN channel on the first network may be optional, i.e. the VPN channel (or multiple VPN channels) may exist simultaneously on the first network and the second network. Furthermore, the VPN channel (or multiple VPN channels) may be simultaneously active on both networks for a short period of time (until successful transition from one network to another network can be confirmed) or for a longer period of time.

[0080] At step **405**, a VPN channel is active between the client and a first network. At optional step **410**, the client detects an access point for a second network (which may or may not be a captive network). At step **415**, the client transmits a signal to the second network access point to determine whether the second network is a captive network. At step **420**, for example, if the second network is indeed a captive network, the captive portal authentication server determines whether the client has already been registered with (and has permission to access) the second network. This may be accomplished, for example, by maintaining a list of the MAC address of any client that is permitted to access the captive network. In the example, at step **420**, the second network determines whether the MAC address of the client is on the list. At step **430**, if access to the second network is restricted (i.e. the client has not been registered with the second network), it is determined that the second network is indeed a captive network.

[0081] In an exemplary embodiment of the present invention, if a captive portal is detected, a user may be alerted (desktop: in app prompt; mobile: notification) and given an option to launch a new browser window (desktop: separate window in app process; mobile: separate browser activity in app) to complete the captive portal login procedure. In another embodiment of the present invention, a program may request the captive portal page, and submit a response to complete the captive portal login procedure with no user interaction being required.

[0082] At step **435**, the captive network access requirements are satisfied. As explained above, the captive network access requirements may vary and may include, for example, agreeing to terms of use, providing an email address, providing a credit card, etc. Processing then proceeds to step **440** (FIG. 5) by off page connector “A”. At step **440**, if the requirements of the captive network have been satisfied, then full (or greater) access to the captive network is granted. At step **445**, the client receives acknowledgment that full (or greater) access to the captive network has been granted. This acknowledgement may be, for example, allowing the client to retrieve and display a web page over the Internet. This acknowledgment may be, for example, sending the client a message that access to the captive portal has been granted. At step **450**, the VPN channel is now active over the second network. Steps **445** and **450** may be interchanged. At step **455**, the VPN channel over the first network is optionally discontinued.

[0083] In a further embodiment of the present invention, a user closing a browser window may trigger a recheck of captive portal status. Alternatively, a captive portal check could be triggered by the failure of a transport socket. As further alternative, a timer in software could trigger the recheck of captive portal status on a regular basis.

[0084] In the above description, the word “access” has been used. It is understood that “access” may refer to the act of transitioning from zero access to an amount of access greater than zero. It is also understood that “access” may refer to the act of transitioning from a first amount of access to a second amount of access greater than the first amount. It is also understood that “access” may refer to the act of transitioning from a) less than full access to a resource to b) full access to that resource.

[0085] In the above description, a “client” has been described that is secure. It is understood that a client may include a communications device and/or computer that is able to transmit and/or receive data in a manner so that it is secured, i.e. data that is not available for inspection unless permitted. Encryption, for example, may render a client and/or data transferred therewith secure.

[0086] The above explanation has included exemplary embodiments to permit a secure channel (such as a VPN channel) that is active on a first network to continue to be active on a second network. One advantage of the above features is that, while using a VPN, when transitioning from a first network to a second network protected by a VPN, the transition is seamless. Otherwise, if transition from a first network to a captive network is attempted during VPN communication, the VPN communication may simply fail at the time when the transition is attempted. Furthermore, if VPN communication is occurring over one network, and an access point to a second network becomes available, the above features enable the VPN communication to then occur in a multipath environment. Otherwise, if the second network is protected by a captive portal, an attempt to continue the VPN communication in a multipath environment may fail, as the captive portal will prevent the establishment of the VPN channel over the captive network.

[0087] In an exemplary embodiment of the present invention a computer system may be included and/or operated within which a set of instructions, for causing the machine to perform any one or more of the methodologies discussed herein, may be executed. In alternative embodiments, the machine may be connected (e.g., networked) to other machines in a local area network (LAN), an intranet, an extranet, or the Internet. The machine may operate in the capacity of a server or a client machine in a client-server network environment, or as a peer machine in a peer-to-peer (or distributed) network environment. The machine may be a personal computer (PC), a tablet PC, a set-top box (STB), a personal digital assistant (PDA), a cellular telephone, a web appliance, a

server, a network router, switch or bridge, or any machine capable of executing a set of instructions (sequential or otherwise) that specify actions to be taken by that machine. Further, while only a single machine is illustrated, the term “machine” shall also be taken to include any collection of machines that individually or jointly execute a set (or multiple sets) of instructions to perform any one or more of the methodologies discussed herein.

[0088] The exemplary computer system includes a processing device, a main memory (e.g., read-only memory (ROM), flash memory, dynamic random access memory (DRAM) (such as synchronous DRAM (SDRAM) or Rambus DRAM (RDRAM), etc.), a static memory (e.g., flash memory, static random access memory (SRAM), etc.), and a data storage device, which communicate with each other via a bus.

[0089] Processing device represents one or more general-purpose processing devices such as a microprocessor, central processing unit, or the like. More particularly, the processing device may be complex instruction set computing (CISC) microprocessor, reduced instruction set computer (RISC) microprocessor, very long instruction word (VLIW) microprocessor, or processor implementing other instruction sets, or processors implementing a combination of instruction sets. Processing device may also be one or more special-purpose processing devices such as an application specific integrated circuit (ASIC), a field programmable gate array (FPGA), a digital signal processor (DSP), network processor, or the like. Processing device is configured to execute listings manager logic for performing the operations and steps discussed herein.

[0090] Computer system may further include a network interface device. Computer system also may include a video display unit (e.g., a liquid crystal display (LCD) or a cathode ray tube (CRT)), an alphanumeric input device (e.g., a keyboard), a cursor control device (e.g., a mouse), and a signal generation device (e.g., a speaker).

[0091] Data storage device may include a machine-readable storage medium (or more specifically a computer-readable storage medium) having one or more sets of instructions embodying any one or more of the methodologies of functions described herein. Data storage may also reside, completely or at least partially, within main memory and/or within processing device during execution thereof by computer system; main memory and processing device also constituting machine-readable storage media.

[0092] Machine-readable storage medium may also be used to store the device queue manager logic persistently. While a non-transitory machine-readable storage medium is shown in an exemplary embodiment to be a single medium, the term “machine-readable storage medium” should be taken to include a single medium or multiple media (e.g., a centralized or distributed database, and/or associated caches and servers) that store the one or more sets of instructions. The term “machine-readable storage medium” shall also be taken to include any medium that is capable of storing or encoding a set of instruction for execution by the machine and that causes the machine to perform any one or more of the methodologies of the present invention. The term “machine-readable storage medium” shall accordingly be taken to include, but not be limited to, solid-state memories, and optical and magnetic media.

[0093] The components and other features described herein can be implemented as discrete hardware components or integrated in the functionality of hardware components such as ASICs, FPGAs, DSPs or similar devices. In addition, these components can be implemented as firmware or functional circuitry within hardware devices. Further, these components can be implemented in any combination of hardware devices and software components.

[0094] Some portions of the detailed descriptions are presented in terms of algorithms and symbolic representations of operations on data bits within a computer memory. These algorithmic descriptions and representations are the means used by those skilled in the data processing arts to most effectively convey the substance of their work to others skilled in the art. An algorithm is here, and generally, conceived to be a self-consistent sequence of steps leading to a desired result. The steps are those requiring physical manipulations of physical quantities. Usually, though not

necessarily, these quantities take the form of electrical or magnetic signals capable of being stored, transferred, combined, compared, and otherwise manipulated. It has proven convenient at times, principally for reasons of common usage, to refer to these signals as bits, values, elements, symbols, characters, terms, numbers, or the like.

[0095] In the aforementioned description, numerous details are set forth. It will be apparent, however, to one skilled in the art, that the disclosure may be practiced without these specific details. In some instances, well-known structures and devices are shown in block diagram form, rather than in detail, in order to avoid obscuring the disclosure.

[0096] The disclosure is related to an apparatus for performing the operations herein. This apparatus may be specially constructed for the required purposes or it may comprise a general-purpose computing device selectively activated or reconfigured by a computer program stored therein. Such a computer program may be stored in a non-transitory computer readable storage medium, such as, but not limited to, any type of disk including floppy disks, optical disks, CD-ROMs and magnetic-optical disks, read-only memories (ROMs), random access memories (RAMs), EPROMs, EEPROMs, magnetic or optical cards, flash memory devices including universal serial bus (USB) storage devices (e.g., USB key devices) or any type of media suitable for storing electronic instructions, each of which may be coupled to a computer system bus.

[0097] Whereas many alterations and modifications of the disclosure will no doubt become apparent to a person of ordinary skill in the art after having read the foregoing description, it is to be understood that any particular implementation shown and described by way of illustration is in no way intended to be considered limiting. Therefore, references to details of various implementations are not intended to limit the scope of the claims, which in themselves recite only those features regarded as the disclosure.

Claims

1. A method comprising: receiving data, via a client device, over a first connection of a first network; sending a request, via the client device, to access a second network detected by the client device while the client device is communicating over the first network; responsive to receiving a captive portal from the second network, forwarding, via the client device, authentication information over a second connection to the second network; and receiving additional data, by the client device, over the first connection provided by the first network and a second connection provided by the second network after the authentication information is approved.
2. The method of claim 1, wherein the additional data is received by the client device concurrently from the first connection and the second connection.
3. The method of claim 1, wherein the additional data received by the client device is received as bonded channels providing unique data packets over each channel.
4. The method of claim 1, wherein the first network is accessible by a first access point and the second network is accessible by a second access point separate from the first access point.
5. The method of claim 1, comprising: forwarding a data request to a virtual private network (VPN) server, via the client device, destined for a remote server to obtain data from the remote server.
6. The method of claim 1, wherein the first communication network is a Wi-Fi network and the second communication network is one or more of a cellular network and another Wi-Fi network.
7. The method of claim 1, wherein the request is sent to an authorization server that authorizes access to the second communication network.
8. A client device comprising: a processor configured to perform receiving data over a first connection of a first network; sending a request to access a second network detected by the client device while the client device is communicating over the first network; responsive to receiving a captive portal from the second network, forwarding authentication information over a second connection to the second network; and receiving additional data over the first connection provided

by the first network and a second connection provided by the second network after the authentication information is approved.

9. The client device of claim 8, wherein the additional data is received by the client device concurrently from the first connection and the second connection.

10. The client device of claim 8, wherein the additional data received by the client device is received as bonded channels providing unique data packets over each channel.

11. The client device of claim 8, wherein the first network is accessible by a first access point and the second network is accessible by a second access point separate from the first access point.

12. The client device of claim 8, wherein the client device is further configured to perform forwarding a data request to a virtual private network (VPN) server destined for a remote server to obtain data from the remote server.

13. The client device of claim 8, wherein the first communication network is a Wi-Fi network and the second communication network is one or more of a cellular network and another Wi-Fi network.

14. The client device of claim 8, wherein the request is sent to an authorization server that authorizes access to the second communication network.

15. A non-transitory computer readable storage medium configured to store instructions that when executed cause a processor to perform: receiving data, via a client device, over a first connection of a first network; sending a request, via the client device, to access a second network detected by the client device while the client device is communicating over the first network; responsive to receiving a captive portal from the second network, forwarding, via the client device, authentication information over a second connection to the second network; and receiving additional data, by the client device, over the first connection provided by the first network and a second connection provided by the second network after the authentication information is approved.

16. The non-transitory computer readable storage medium of claim 15, wherein the additional data is received by the client device concurrently from the first connection and the second connection.

17. The non-transitory computer readable storage medium of claim 15, wherein the additional data received by the client device is received as bonded channels providing unique data packets over each channel.

18. The non-transitory computer readable storage medium of claim 15, wherein the first network is accessible by a first access point and the second network is accessible by a second access point separate from the first access point.

19. The non-transitory computer readable storage medium of claim 15, wherein the processor is further configured to perform: forwarding a data request to a virtual private network (VPN) server, via the client device, destined for a remote server to obtain data from the remote server.

20. The non-transitory computer readable storage medium of claim 15, wherein the first communication network is a Wi-Fi network and the second communication network is one or more of a cellular network and another Wi-Fi network.
